

EUROPEAN ECONOMICS ANALYST

This Time is Different (from 2022) — The Impact of Higher Energy Prices on European Manufacturing Industries

- The conflict in the Middle East and the near closure of the Strait of Hormuz (SoH) continue to cause large-scale disruptions to global energy supply with meaningful increases in energy prices over the last month and a notable upward revision to our energy price forecasts. In this week's Analyst, we discuss the implications of the energy shock across European manufacturing industries, highlighting three key differences vs the 2022/23 European energy crisis.
- First, the current price shock is expected to be smaller and less persistent, while the decoupling from Russian pipeline gas that began in 2022/23 has been sustained. Accordingly, we expect the aggregate impact of the shock to be smaller, and capacity shutdowns in energy-intensive industries should be less likely than during the 2022/23 crisis. We estimate a 2% hit to industrial production by the end of 2027, which compares to our previous estimate of a 4% hit to industrial production from the 2022/23 crisis.
- Second, the current supply shock is less gas-centric and more oil-driven. The 2022/23 crisis was fundamentally driven by a sharp reduction in Russian pipeline gas supply to Europe and impacted a narrow set of highly gas-intensive industries particularly hard. The current shock is primarily an oil supply one that tends to reduce industrial production more broadly while also impacting capital goods producers and export-oriented sectors negatively.
- Third, the current shock is far less Europe-specific and more global in nature. In 2022/23, energy market shifts were largely focused on Europe, leading to major cost disadvantages for energy-intensive industries, particularly relative to China, and a surge in import competition. This time, energy price pressures are more global, with Asia appearing to be affected to a similar (or even greater) extent. European energy-intensive industries therefore look less exposed to competitiveness pressures. Non-energy intensive industries—that previously substituted more expensive domestic energy-intensive intermediates with lower cost imports—are likely to be relatively more exposed to the current shock.
- Taken together, our analysis suggests that the overall impact of the current energy price shock on European manufacturing should be notably smaller than in 2022, with the shock expected to be less severe and less persistent. The energy shock is also likely to be less concentrated in energy-intensive industries than during the European energy crisis, but it may weigh relatively more on the broader manufacturing sector, as it is more oil-focused and may have a larger

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impact on global demand.

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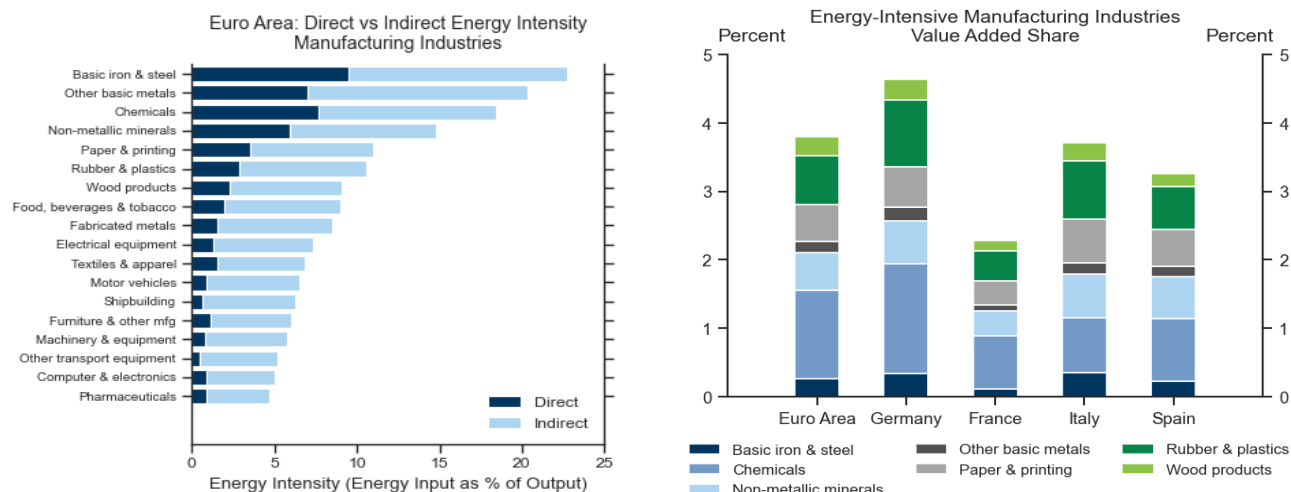
The conflict in the Middle East and the near closure of the Strait of Hormuz (SoH) continue to cause large-scale disruptions to global energy supply. Energy prices have increased substantially since late February and our commodities analysts raised their forecasts significantly, expecting Brent oil at \$83/bbl on average in 2026 (vs. \$64/bbl before the conflict) and European natural gas (TTF) at EUR44/MWh (vs EUR34/MWh before).

In this week's Analyst, we analyse the implications of the energy price surge for European manufacturing industries, drawing on lessons from the 2022/23 European energy crisis and highlighting key differences between the current and the previous energy supply shocks.

Heterogeneous Impact Across Industries, Depending on Their Energy Intensity

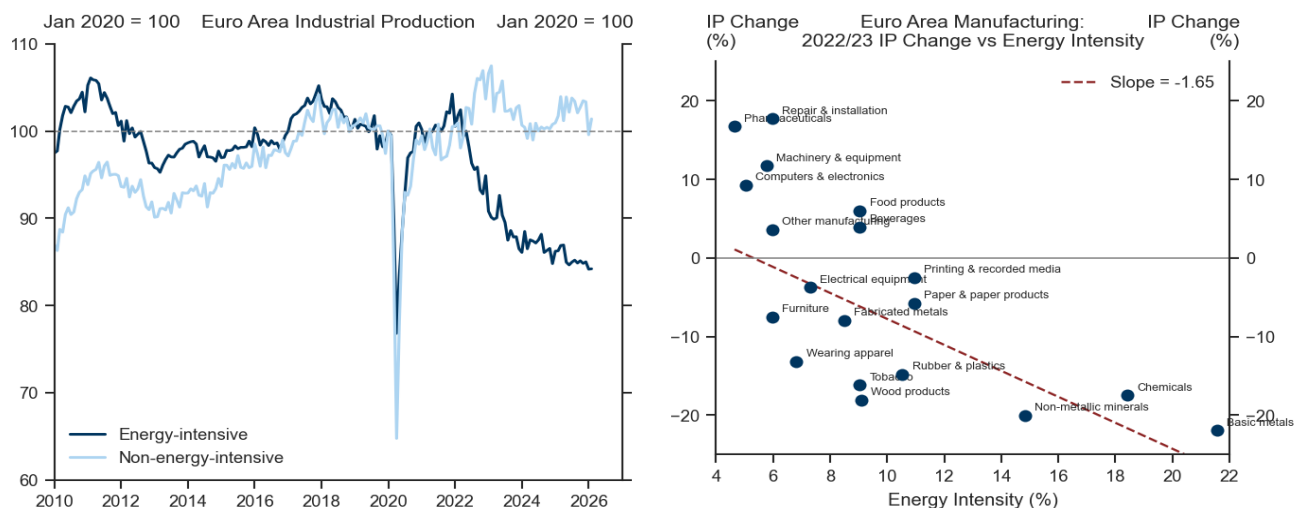
Energy price increases raise production costs through higher intermediate input prices, both directly for energy inputs and indirectly for energy-intensive intermediates. Energy intensity—measured as the direct and indirect value of energy inputs relative to total output—varies substantially across industries ([Exhibit 1](#)).¹ Across countries, the value-added share of energy-intensive industries—excluding energy production itself—differs substantially, although this is mostly driven by the variation in the overall manufacturing share.

¹ We use the OECD Intercountry Input Output tables to calculate direct and indirect energy inputs for each industry. We classify oil (NACE code B5) and gas (B6) extraction, coke and refining (C19) and electricity, gas, water and steam supply (D) as direct energy inputs. Indirect energy inputs are recovered through the Leontief inverse.

Exhibit 1: Energy Intensity Varies Widely Across Industries

Source: Haver Analytics, Goldman Sachs Global Investment Research, OECD

Energy-intensive industries were hit hard during the 2022/23 European energy crisis. While production in non-energy intensive industries continued to grow until early-2023 and declined only slightly during 2023, energy-intensive industries on aggregate reduced production by around 15% between 2022 and 2024 (Exhibit 2). Across industries, energy intensity was highly correlated with production losses, which might suggest that these industries would also be most strongly affected by the current energy price shock. That said, the current shock differs from the 2022/23 European energy crisis along three dimensions.

Exhibit 2: Industrial Production in Energy Intensive Industries Declined by 15% During the Energy Crisis

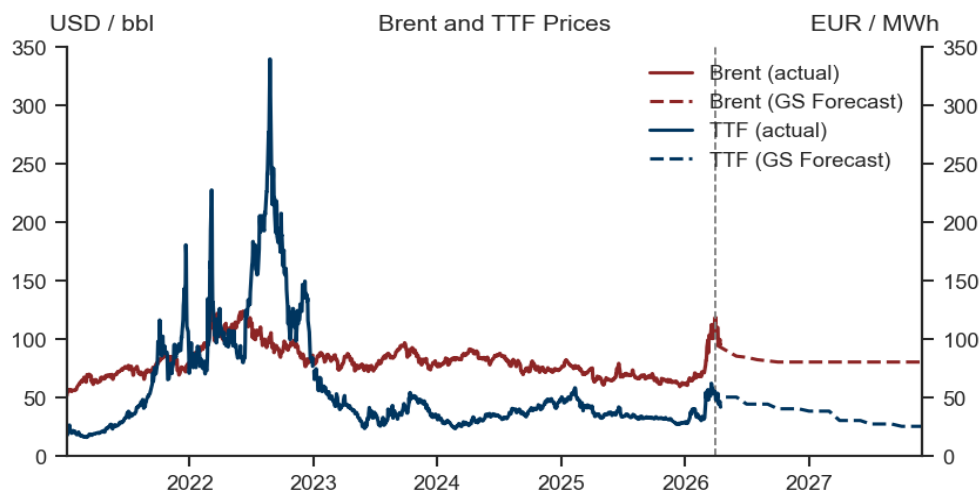
Source: Haver Analytics, Goldman Sachs Global Investment Research, OECD

Difference 1: The Current Shock is Likely to be Smaller and Less Persistent than the 2022/23 Energy Crisis

First, the energy price shock has been much smaller so far and is expected to be more transitory than in 2022/23. Brent oil prices averaged \$99/bbl in 2022, up around 40%

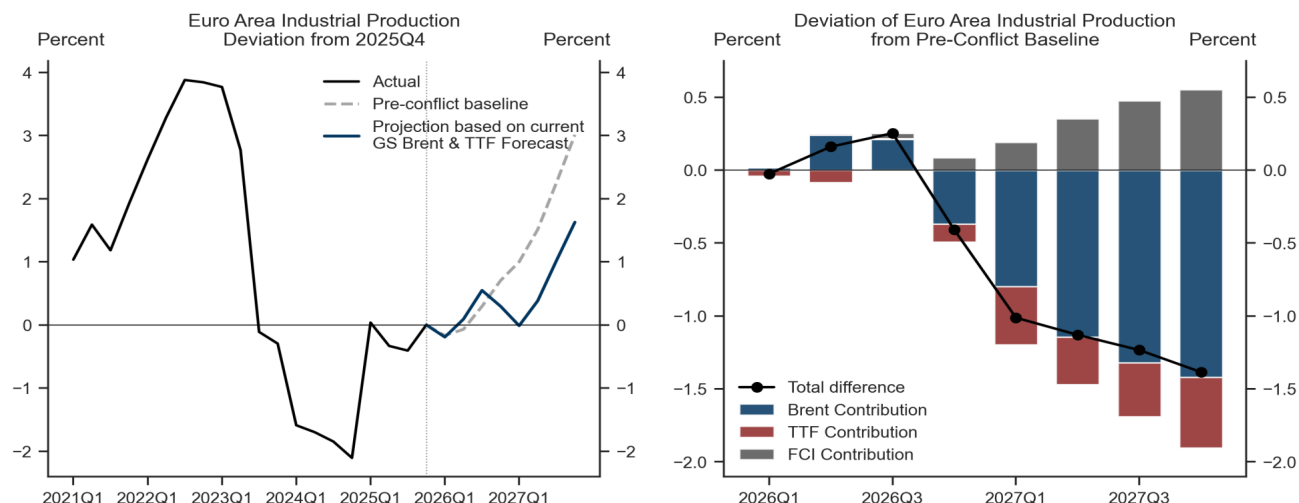
from 2021, and European natural gas prices averaged EUR133/MWh, up around 180% from 2021; this compares to a roughly 20% to 30% price increase between the 2025 and 2026 annual averages conditional on our forecasts ([Exhibit 3](#)). In addition, the current shock is expected to be much less persistent than the European energy crisis, while the loss of European gas supply via Russian pipelines was seen as sustained. Accordingly, the risk of plant closures and permanent capacity reduction in energy-intensive industries such as chemicals, which resulted in a decline in the non-cyclical component of industrial production in 2022/23, should be meaningfully lower this time.

Exhibit 3: For Europe, the Current Energy Price Shock is Likely to be Smaller and More Transitory than the 2022/23 Energy Crisis



Source: Bloomberg, Haver Analytics, Goldman Sachs Global Investment Research

We quantify the potential impact on overall industrial production relative to our pre-conflict baseline using a dynamic panel model we have employed before to gauge the impact of the 2022/23 energy crisis. We condition our forecast on the price paths expected by our commodities strategists and assume constant financial conditions. The model projects that higher oil and gas prices lower Euro Area industrial production by almost 2% by end-2027 relative to our pre-conflict baseline, with negative IP growth later this year but a return to positive growth in 2027. This compares to an estimated drag of 4% during the energy crisis of 2022/23.

Exhibit 4: We Expect the Current Energy Shock to Lower Euro Area Industrial Production by 1.5% by 2027Q4

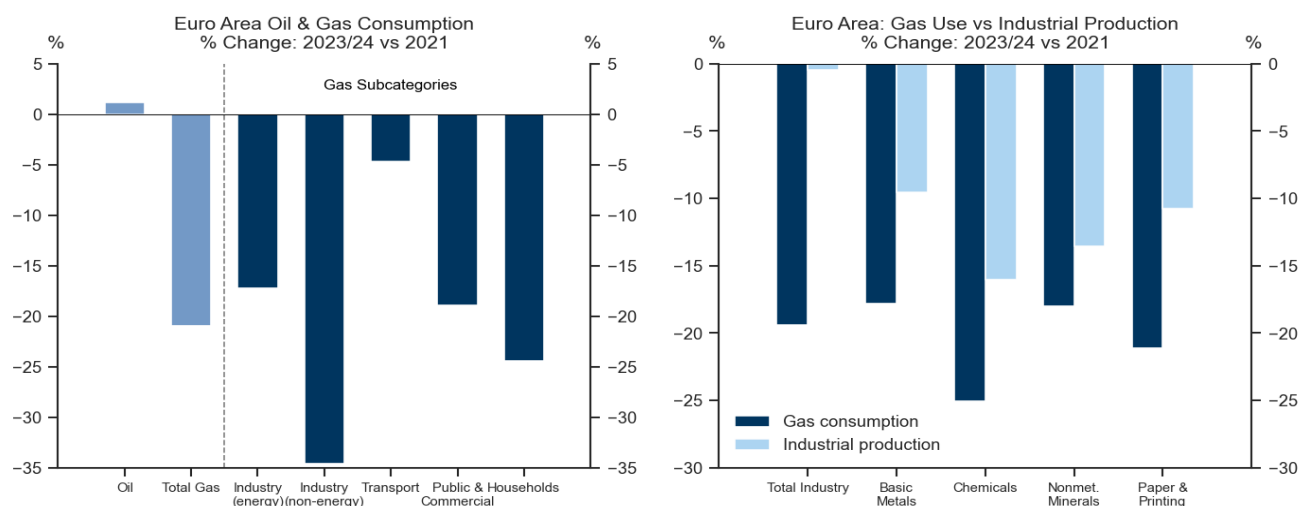
Source: Haver Analytics, Goldman Sachs Global Investment Research

Difference 2: The Current Shock is Less Centred on Gas and More on Oil

Second, the European energy crisis was primarily concentrated in natural gas, while the current episode appears to affect global oil supply more than global natural gas supply. Natural gas supply in Europe declined by about 20% between 2021 and 2023/24, and the consumption response was relatively evenly spread across industry, services and households ([Exhibit 5](#), left). Oil consumption remained flat compared to 2021 levels.

Our commodities team currently estimate a roughly 8% hit to global oil production in March, and a hit to Persian Gulf exports of about 12% of global oil supply. While about 20% of global LNG supply that typically passes through the Strait of Hormuz is currently blocked, LNG accounts for only about 15% of global gas supply and about 45% of EU gas supply, and the hit to EU gas supply would therefore be meaningfully lower than in 2022/23.² Furthermore, our commodities analysts expect LNG supply to increase substantially over the coming years.

² While our commodities analysts initially expected Europe having to deliver most of the demand response to the LNG supply shock, they recently noted that China reduced its net LNG demand notably, reducing the supply shock to Europe.

Exhibit 5: The 2022/23 European Energy Crisis was Concentrated in Natural Gas, and Gas Savings in Industry Were Tightly Connected to Production Cuts


Source: Eurostat, Haver Analytics, Goldman Sachs Global Investment Research

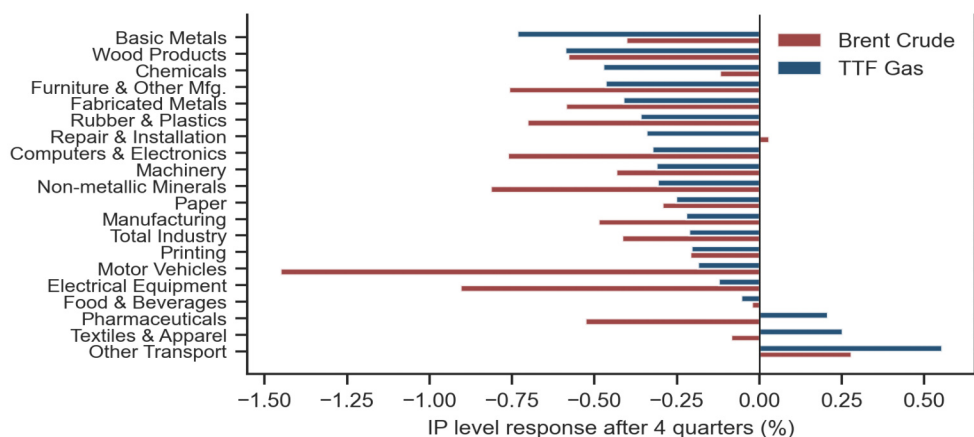
The industry response to the 2022/23 energy crisis was driven by four gas-intensive industries, which together accounted for about 70% of industrial gas consumption ([Exhibit 5, right](#)). In these industries, reduced gas consumption also led to lower production, although less than proportional to the reduction in gas consumption, indicating efficiency gains and production cuts in particularly gas-intensive goods. Aggregate industrial production stayed roughly flat, as output losses in gas-intensive industries—which accounted for about 18% of manufacturing output—were offset by gains in non-gas intensive industries.

Leveraging a panel of sector-level industrial production across the major Euro Area economies, we find that supply-driven gas price shocks most negatively impact production in energy-intensive industries like basic metals or chemicals ([Exhibit 6](#)). While oil price shocks also weigh on energy-intensive industries, they affect manufacturing more broadly. Industrial production in capital goods producing industries, such as motor vehicles, machinery and equipment, computers & electronics, and electrical equipment, tends to decline notably after oil supply shocks.

A key reason for the different impact is likely that gas markets are less globally integrated than oil markets, even less so in the past. Oil supply shocks thus tend to have more negative consequences for global demand. Accordingly, while the current shock is expected to be smaller and less persistent than the 2022/23 energy crisis, it should be less concentrated in energy-intensive industries and weigh relatively more on export-oriented industries.

Exhibit 6: The Negative Effect of Gas Price Shocks is Concentrated in Energy Intensive Industries; Oil Price Shocks Typically Have Broader Negative Effects

Industrial Production Response to 10% Energy Price Shock After 4Q

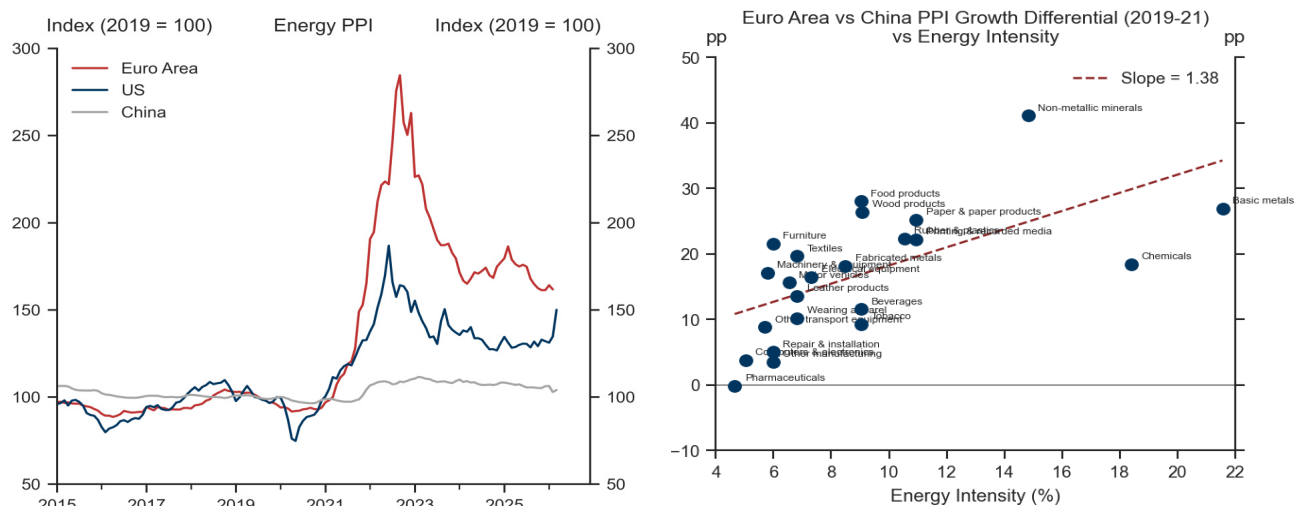


Source: Goldman Sachs Global Investment Research, Haver Analytics

Difference 3: The Current Shock is Less Concentrated on Europe

Third, the energy price shock in 2022 was geographically asymmetric, while the current shock affects Asia to a similar (or even higher degree) compared to Europe. Broadly, energy prices in Europe increased by almost 200% between early 2021 and mid-2022, while energy prices in the US increased by less than 100% and energy prices in China by about 10% ([Exhibit 7, left](#)). European energy prices remained about 75% above their pre-crisis levels throughout 2023 and 2024, while they remained only about 25% higher in the US and only about 10% higher in China. This asymmetric energy price shock translated into substantial cost disadvantages for energy-intensive industries in Europe, in particular relative to China. The increase in European producer prices relative to China was much higher in energy-intensive than in non-energy intensive industries ([Exhibit 7, right](#)).

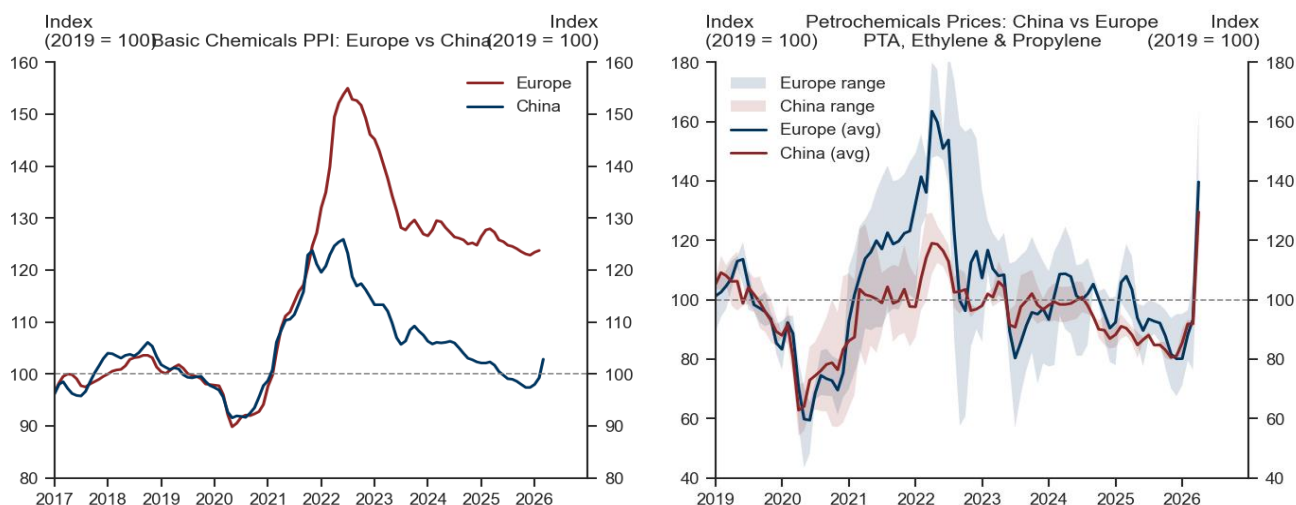
Exhibit 7: The Energy Price Shock of 2022 was Very Asymmetric and Lowered the Competitiveness of European Energy-Intensive Industries



Source: Haver Analytics, Goldman Sachs Global Investment Research, OECD

Early indicators point to a much less asymmetric shock this time. Producer prices for basic chemicals have started to increase meaningfully in China and petrochemicals prices in China and Europe have risen to a similar degree over the past month ([Exhibit 8](#)).

Exhibit 8: Price Pressures in Europe and China are More Similar Than in 2022



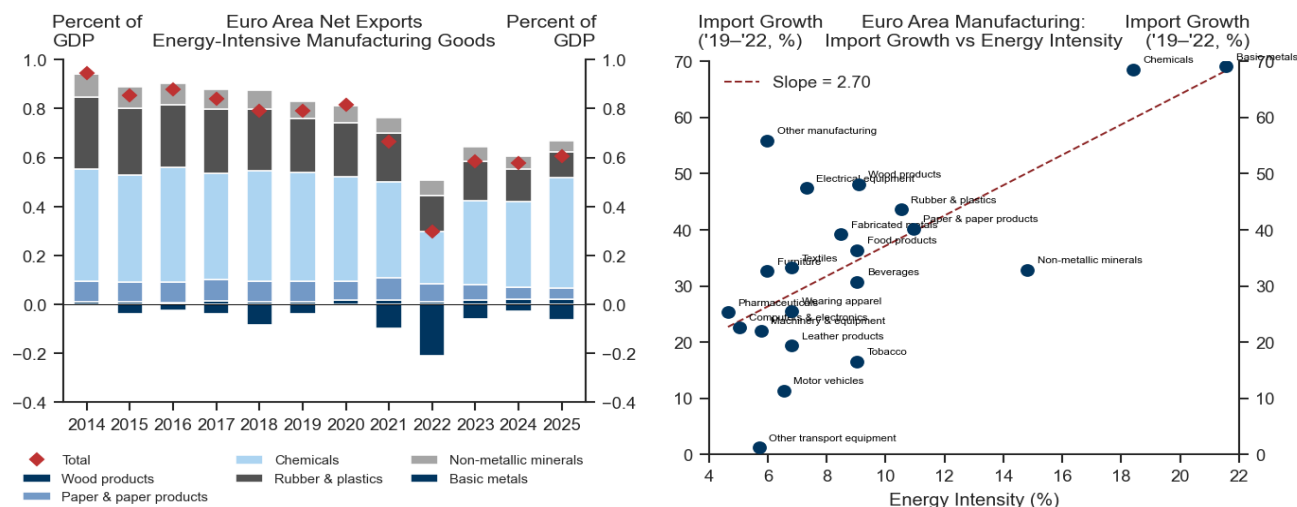
LHS: Last observation for China is March 2026, last observation for the Euro Area is February 2026.

Source: Haver Analytics, Goldman Sachs Global Investment Research, Bloomberg

The loss in competitiveness resulted in a decline in net exports, by about half in nominal terms in 2022 and as a share of GDP ([Exhibit 9](#), left). This was driven by a particularly strong increase in imports in the most energy-intensive industries ([Exhibit 9](#), right). This surge in imports, on the one hand, intensified competition in energy-intensive industries and weighed on production. On the other hand, lower-cost imports from countries less affected by energy price increases helped reduce cost pressures for industries that use energy-intensive intermediates as inputs. Import substitution thus magnified the negative impact on energy-intensive industries but cushioned less energy-intensive

industries and prevented “cascading effects” from production shortfalls in energy-intensive sectors. Under the current—more global and less asymmetric—energy shock, we expect energy-intensive industries to fare relatively better and non-energy intensive industries relatively worse than in 2022/23.

Exhibit 9: Net Exports of Energy-Intensive Manufacturing Declined Substantially in 2022, Driven by a Surge in Imports



Source: Haver Analytics, Goldman Sachs Global Investment Research

Manufacturing Drag Should be Notably Smaller than in 2022/23, and Less Concentrated in Energy-Intensive Industries

Taken together, our analysis suggests that the overall impact of the current energy price shock on European manufacturing should be notably smaller than in 2022, with the shock expected to be less severe and less persistent. The energy shock is also likely to be less concentrated in energy-intensive industries than during the European energy crisis, but it may weigh relatively more on the broader manufacturing sector, as it is more oil-focused and may have a larger impact on global demand.

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